Session 16 · Tuesday 20 October 2026 · The occupancy derivation and the state count are projected one line at a time; the board carries the counting, the depths, and the two conditions.

Board notes

Print this and carry it. The deck records what is *projected*; this records what gets *written*.

Four board moments, named by the deck's segment rather than by the clock: the four-state count at **nor-build**, the two XNOR networks at **nor-debrief**, the two conditions at **fix-separate**, and $S(n)$ at **dna-count**. Nothing here is on a slide except as a finished line.

Conventions. ASK: put it to the room before writing, and the answer to expect. **CHECK**: the sanity check to do out loud, because doing them is the habit being taught. **IF ASKED**: the question that always comes. **CUT**: what to drop if the hour is going long, chosen now rather than later.

Before class — left wing, and leave it up

Two things, and they stay up the whole period.

The tick rule.

> *every gene reads its inputs **as they were last tick**, all genes update together, and at $t = 0$ every gene is at 0*

The three kinds.

> ***static-0** — should have stayed 0, went up and came back **static-1** — should have stayed 1, went down and came back **dynamic** — should have changed once, changed more than once*

Say once, and do not belabour it: only two of those can happen in a cell that has just been made. From an empty cell every gene starts at 0, so the output's first move is always upward — a **static-1 needs a circuit that is already running**, which is what handout item 3(d) has them do. It is also why, in a cold start, the kind carries no information the settled value did not already give you: settles to 0 and it is static-0, settles to 1 and it is dynamic, every time.

The rest of the board stays empty. It fills at **nor-build**.

Centre, at nor-build — where the gate comes from

This is the derivation the projected run shows one line at a time. Do it on the board **first**, then advance the slides over it; the slides are the record, the board is the rate.

Two operator sites on one promoter. Four states, and their weights relative to the empty promoter:

$$1, \qquad r_A, \qquad r_B, \qquad r_A r_B \qquad \text{where } r = [\text{repressor}]/K$$

ASK before writing the next line: *what do we do with four weights?* Add them. They came off this in session 6 and most of the room will say it.

$$Z = 1 + r_A + r_B + r_A r_B = (1 + r_A)(1 + r_B)$$

CHECK out loud: it factorises. Say why that is not a coincidence — the two sites are independent, so the partition function of the pair is the product of the two one-site partition functions. If they are *not* independent, it does not factorise, and that is cooperativity.

Only the empty promoter fires, so

$$P_{\text{on}} = \frac{1}{(1 + r_A)(1 + r_B)}$$

ASK: *push r_A to zero and to infinity, and the same for r_B. Four corners. What are they?* $1, 0, 0, 0$.

Write underneath, and leave it:

> ***that is NOR, and nobody chose it — the DNA layout wrote it***

IF ASKED *what if the two sites overlap?* Then there is no doubly-bound state, $Z = 1 + r_A + r_B$, and the gate is still NOR because the empty state is still the only one that fires. Good question, and the answer is that NOR is robust to that detail.

Right wing, at nor-debrief — the two networks, and the counting

Take the networks the room actually produced and sort them against the two on the slide. Then do the counting on the board, because the counting is where the error is.

Write the recipe network as a list:

> $\neg A$, $\neg B$, $P = \text{NOR}(A, B)$, $Q = \text{NOR}(\neg A, \neg B)$, $R = \text{NOR}(P, Q)$, *out* $= \text{NOT } R$

ASK: *how deep is it?* Expect "three". Then trace two routes with a finger, out loud:

> $A \to P \to R \to$ *out* — ***three*** $A \to \neg A \to Q \to R \to$ *out* — ***four***

Four and three, not three and two. Say that the number people reach for is the number of *layers in the list*, and the number that matters is the number of *genes on a route*. This is the one place in the session where the room will write down a wrong number and not notice.

Then the minimum, four gates with paths of 3 and 2. Do not derive it — say that an exhaustive search over two-input NOR networks finds it and that the search is the compiler's job.

Leave on the board, and do not explain it:

> ***the paths are not the same length — in both of them***

The hazard segment collects on that line. It is much stronger if they wrote the numbers themselves.

Centre, at fix-separate — what actually decides it

The slide has the computed table. The board has the one line it comes down to. Write the line **before** the table goes up, and leave it:

> ***an absent activator holds a gene off — an absent repressor lets it fire***

ASK before showing the slide: *we refuted "balance is enough". Is "OFF when empty" enough?* Most of the room will say yes, because it is the answer the previous two slides set up. It is not, and the slide has the counterexample: take $Z = \mathrm{NOR}(\neg A, \neg B)$ — already balanced — and bolt an activator-driven output onto it. Now it is OFF when empty **and** balanced, and it still glitches three times in four.

Then the row that does the other half of the work, and it is worth slowing down for:

> $g = \mathrm{BUF}(A)$, *out = a promoter needing* g, *blocked by* B *— two genes,* ***unbalanced*** *(A in two, B in one),* ***clean in all four****, and it computes* $A \wedge \neg B$*, the classifier we have failed to build all hour*

ASK: *why does this one get away with unequal legs?* Because the late signal is the **activator**. Write the condition as the inequality it is:

$$\mathrm{depth}(\mathrm{repressors}) \;\leq\; \mathrm{depth}(\mathrm{activators})$$

CHECK out loud: a repressor arriving *early* is harmless — the gate is held off either way. Equal depth is the special case, not the condition. That is the whole reason balance looked like the answer for two slides.

IF ASKED *so can I just put an activator on the output of anything?* No, and (ii) is why: Z is a NOR, so Z is already **wrong**, and the BUF relays the wrong value one tick later. The condition has to hold at every gene on the route, not only the last. You cannot repair a circuit at its output.

IF ASKED *is the last column a theorem?* It is a **sufficient** condition — it refuses some circuits that are in fact clean — checked by exhaustion over every two- and three-gate network in this model (`cold_start_safe`, `tests/test_logic.py`). Every gate here has one unit of delay and real promoters do not. Say that; it is ten seconds and it is the honest end of the segment.

> ***This surface has been wrong twice.*** *First "unequal paths cause it", refuted by Z. Then "fix the promoter, not the wiring", refuted by the padded repair. Then "you need both", refuted by the two rows above. If you find yourself about to state a new sufficient condition at the board, don't — point at the computed column instead.*

Right wing, at dna-count — counting the orders

Three lines, and it is worth doing them rather than quoting the formula.

ASK: *what does a state of a staggered device record?* Which invertases have fired, **and in what order**.

So a state is an ordered sequence drawn from a subset of the n available:

$$\underbrace{\binom{n}{k}}_{\text{which } k} \times \underbrace{k!}_{\text{in what order}} \qquad \text{summed over } k = 1 \ldots n$$

$$S(n) = \sum_{k=1}^{n} k! \binom{n}{k}$$

CHECK at $n = 2$: $2 + 2 = 4$, plus the starting state is five — and five is what Ham's Fig. 1C draws, which is on the screen two slides earlier. Do this check. It is the only thing standing between this formula and a number nobody can verify.

$n = 3$ gives 15, against 8 for three independent bits. $n = 10$ gives nearly 10^7, against 1024.

IF ASKED *why doesn't nesting do this?* Because nested inversions commute: firing a then b lands on the same arrangement as b then a, so the order is not recoverable and there is nothing extra to count.

IF ASKED *what about firing the same invertase twice?* Good question and the honest answer is that Ham's count does not include it — a state is a sequence of *distinct* invertases. Whether a real device reaches more arrangements than that is open, and it is in `DESIGN.md` §8.5 rather than on a slide.

CUT, in this order

1. **dna-count** at the board — quote $S(n)$ from the slide and keep only the $n = 2$ check against Ham's five-state graph.
2. The **IF ASKED** on overlapping operator sites at nor-build.
3. The inequality at **fix-separate** — the computed slide carries the verdict column; the board version is for the room that wants to see why a late activator is safe and a late repressor is not. Keep the one-line rule even if you cut the derivation of it.

Do **not** cut the counting at nor-debrief. The wrong number is the one thing on this sheet that the room will otherwise carry out of the building.